V.L Memorial Public School

475 Vijay Nagar

CHAPTER-10-QUESTIONS-CLASS8TH-SCIENCE

Class 08 - Science (Curiosity)

Section A

1. How does a concave lens make an object appear? **[1]**

   a) Inverted and enlarged
   b) Erect and enlarged
   c) Erect and diminished
   d) Inverted and diminished

2. Why does the inner side of a spoon behave like a concave mirror? **[1]**

   a) Its surface remains straight.
   b) Its surface curves outward.
   c) Its surface curves inward.
   d) Its surface remains flat.

3. Which mirror is used as the main mirror in most reflecting telescopes? **[1]**

   a) Spherical plane mirror
   b) Convex mirror
   c) Concave mirror
   d) Plane mirror

4. Why are convex mirrors useful at road intersections? **[1]**

   a) They produce inverted images
   b) They maintain object size
   c) They enlarge nearby objects
   d) They provide wider visibility

5. What happens to parallel rays falling on a convex mirror? **[1]**

   a) They remain parallel
   b) They focus at a point
   c) They diverge
   d) They converge

6. Which object can act like a curved mirror when its shiny surface is viewed? **[1]**

   a) Metallic spoon
   b) Wooden board
   c) Flat glass
   d) Paper sheet

7. Which lens makes an object appear diminished in size? **[1]**

   a) Flat plastic strip
   b) Flat glass plate
   c) Convex lens
   d) Concave lens

8. What happens when an object is moved away from a concave mirror? **[1]**

   a) The image becomes larger and inverted
   b) The image becomes inverted and smaller
   c) The image becomes larger and erect
   d) The image remains the same size

9. Which set of rays lies in the same plane during reflection? **[1]**

   a) Incident ray, torch, mirror
   b) Reflected ray, paper, stand

c) Incident ray, normal, reflected ray
d) Normal, torch, paper

10. Why does a concave mirror concentrate sunlight onto a small spot? **[1]**

a) Reflected rays converge
b) Reflected rays diverge
c) Reflected rays remain parallel
d) Reflected rays disappear

11. What happens to a convex-lens image after it first appears enlarged? **[1]**

a) It later diminishes
b) It later brightens
c) It later disappears
d) It later becomes transparent

12. Which lens is used to correct myopia (short-sightedness)? **[1]**

a) Convex lens
b) Plane mirror
c) Concave lens
d) Prism

13. Why does a paper burn when a concave mirror concentrates sunlight on it? **[1]**

a) Due to concentrated heat at the focus
b) Because paper is sensitive to light
c) Due to sunlight's chemical properties
d) Because of increased light speed

14. If you need a mirror to show an enlarged image, which will you use? **[1]**

a) None of these
b) Plane mirror
c) Convex mirror
d) Concave mirror

15. If the angle of incidence is 35 degrees, what is the angle of reflection? **[1]**

a) 55 degrees
b) 25 degrees
c) 70 degrees
d) 35 degrees

16. What type of image is formed when an object is placed close to a concave mirror? **[1]**

a) Absent from mirror
b) Larger than object
c) Smaller than object
d) Equal to object

17. What happens to the image through a convex lens as the object moves farther away? **[1]**

a) It becomes circular
b) It becomes inverted
c) It becomes stationary
d) It becomes transparent

18. Which surface of a spoon acts like a convex mirror? **[1]**

a) Handle
b) Inner surface
c) Outer surface
d) Back side

19. Which type of mirror has a reflecting surface that curves inward? **[1]**

a) Convex mirror
b) Cylindrical mirror
c) Plane mirror
d) Concave mirror

20. At what angle is the normal drawn to the mirror surface? **[1]**

a) 60 degrees
b) 45 degrees

c) 90 degrees
d) 180 degrees

21. **Assertion (A):** When Meena looked into a spherical mirror, her reflection appeared different from that in a plane mirror. **[1]**
Reason (R): Spherical mirrors have curved surfaces that alter the path of light, affecting the appearance of the reflected image.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

22. **Assertion (A):** Convex mirrors can be used as rearview mirrors in vehicles. **[1]**
Reason (R): Convex mirrors provide a wider field of view than plane mirrors.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

23. **Assertion (A):** The image in a concave mirror changes size and orientation based on the object's distance. **[1]**
Reason (R): Concave mirrors reflect light rays that converge at a focal point.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

24. **Assertion (A):** Magnifying glasses enlarge the appearance of small text. **[1]**
Reason (R): Convex lenses enlarge images through focus and curvature.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

25. **Assertion (A):** Lenses allow light to pass through them. **[1]**
Reason (R): Lenses are opaque objects that do not permit the transmission of any light.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

26. **Assertion (A):** A concave lens always produces a virtual image. **[1]**
Reason (R): The nature of an image depends solely on the distance of the object from the lens.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

27. **Assertion (A):** A convex lens is thicker at the middle compared to the edges. **[1]**
Reason (R): A concave lens also thickens at the middle, focusing light similarly to a convex lens.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

28. How are spherical mirrors made? **[1]**

29. Where can convex mirrors commonly be found in everyday surroundings? **[1]**

30. How images of objects appear when viewed through a water drop. **[1]**

31. In general terms, how does a concave mirror differ from a plane mirror regarding beam behavior? **[1]**

32. Why are concave mirrors used in torches and vehicle headlights? **[1]**

33. Explain why spherical mirrors do not represent a real section of a hollow glass sphere. **[1]**

34. Differentiate between the reflection of a parallel beam of light on concave and convex mirrors. **[1]**

Section B

35. Identify two uses of concave mirrors in real life. **[2]**

36. How would you analyze the image changes when using both inner and outer sides of a spoon as mirrors? **[2]**

37. Describe a concave mirror highlighting its characteristics based on its reflecting surface. **[2]**

38. Define a lens and describe its two main types. **[2]**

39. What are the conceptual differences between how a plane mirror and a spherical mirror form images? **[2]**

40. Describe the role of convex mirrors in road intersections. **[2]**

Section C

41. Discuss how spherical mirrors are used in everyday applications, giving specific examples and their function. **[3]**

42. Why do two incident rays striking the same point have normals with the same direction? **[3]**

43. What does bending and flattening the chart paper demonstrate about the plane of reflected light? **[3]**

44. In what ways does the distance of an object from a concave lens affect its appearance? **[3]**

45. What practical applications require the use of convex lenses to magnify images? **[3]**

46. Explain how a convex lens functions when an object is placed at varying distances. **[3]**

47. Explain how the properties of convex mirrors make them useful in vehicles and big stores. **[3]**

48. Differentiate between a convex lens and a concave lens based on their shape. **[3]**

Section D

49. **Read the following text carefully and answer the questions that follow:** **[4]**

The laws of reflection state that the angle of incidence is always equal to the angle of reflection, and the incident ray, reflected ray, and the normal at the point of incidence all lie in the same plane. These laws are universal and apply to all types of mirrors, whether plane or spherical. While the direction of reflected rays may differ with mirror shape, the fundamental laws of reflection always hold true.

Questions:

a. What is one law of reflection? **(1)**

b. Explain the second law of reflection. **(1)**

c. Why are the laws of reflection considered universal? **(2)**

OR

How do the laws of reflection explain the working of spherical mirrors? **(2)**

50. **Read the following text carefully and answer the questions that follow:** **[4]**

Spherical mirrors are curved mirrors that reflect light differently from plane mirrors. They are shaped like parts of a hollow sphere and can be of two types: concave and convex. A concave mirror has an inwardly curved reflecting surface, while a convex mirror has an outwardly bulging reflecting surface. Concave mirrors form inverted or magnified images depending on the position of the object, whereas convex mirrors always form smaller, erect images.

Questions:

a. Which type of spherical mirror always forms a smaller, erect image? **(1)**

b. Differentiate between concave and convex mirrors. **(1)**

c. Explain how concave mirrors form different types of imges. **(2)**

OR

Why are convex mirrors useful as rear-view mirrors? **(2)**

51. **Read the following text carefully and answer the questions that follow:** **[4]**

A lens is a transparent material, usually glass or plastic, with curved surfaces. Lenses can be convex or concave. A convex lens is thicker in the middle and thinner at the edges, while a concave lens is thinner in the middle and thicker at the edges. Unlike mirrors, lenses allow light to pass through them. A convex lens converges light and is called a converging lens, while a concave lens diverges light and is called a diverging lens.

Questions:

a. Which type of lens is also called a converging lens? **(1)**

b. Differentiate between convex and concave lenses based on their shapes. **(1)**

c. Why is a convex lens called a converging lens and a concave lens a diverging lens? **(2)**

OR

Compare the basic function of mirrors and lenses in image formation. **(2)**

Section E

52. Describe the behavior of images formed by a concave mirror as the object is moved closer or farther from the mirror. How do these images differ from those formed by a convex mirror? **[5]**

53. Explain the role of the two laws of reflection when light strikes spherical mirrors. **[5]**

54. How do concave and convex mirrors differently direct parallel rays of light? Explain why. **[5]**

55. Apply the principles of spherical mirrors to explain why a metallic spoon can act as both a concave and convex mirror. **[5]**

56. Evaluate the role of the normal in describing reflection. **[5]**

57. Compare the practical uses of concave and convex mirrors, evaluating their effectiveness in different situations. **[5]**

Section F

58. **State whether the given statement is True or False:** **[8]**

(a) Mirrors help in the refraction of light. **[1]**

(b) A spherical mirror with an inward-curved reflecting surface is called a convex mirror. **[1]**

(c) A concave mirror is also known as a converging mirror. **[1]**

(d) Lenses redirect light by bending it. **[1]**

(e) Light always travels in a straight line in a transparent medium. **[1]**

(f) The light passing through a convex lens is parallel. **[1]**

(g) A concave lens is thicker in the middle than at the edges. **[1]**

(h) The shape of the water drop surface is curved outwards. **[1]**

Section G

59. **Fill in the blanks:** **[10]**

(a) The ray of light that comes back from the mirror is called the _____ ray. **[1]**

(b) A concave lens is a _____ lens because it spreads light rays apart. **[1]**

(c) A periscope uses _____ to reflect light and allow observation around corners or obstacles. **[1]**

(d) A concave lens is also known as a _____ lens. **[1]**

(e) The image on the outer side of a spoon is erect but _____ in size. **[1]**

(f) The angle between the normal and the incident ray is called the angle of _____. **[1]**

(g) The human eye contains a _____ lens that changes shape to focus. **[1]**

(h) A _____ lens is thinner at the middle than at the edges. **[1]**

(i) Using a magnifying glass makes letters appear _____. **[1]**

(j) The human eye uses a _____ lens to form an image on the retina. **[1]**